MEGGIE CHEN

Ithaca, NY 14853 | mc2894@cornell.edu | (650) 224-3189 | www.linkedin.com/in/meggierchen/

EDUCATION

Cornell University, Ithaca NY **Expected December 2027**

B.S. in Mechanical Engineering, GPA: 3.98

Relevant coursework: Statics & Mechanics of Solids, Mechanics of Engineering Materials, Dynamics, Flight Dynamics, System Dynamics, Mechatronics, Thermodynamics, Heat Transfer, Fluid Mechanics, Spacecraft Dynamics

TECHNICAL & LEADERSHIP EXPERIENCE

INCOMING ENGINEERING INTERN @ SPACEX **Spring 2027**

- Supporting the development of ground support and pad infrastructure for Starship on the Launch Mechanisms and Structures team.

CORNELL UNIVERSITY UNMANNED AIR SYSTEMS **November 2024 – Present**

Structures and Payloads Lead

- Lead a 10-engineer structures and payloads subgroup on a 35 lb autonomous search-and-rescue VTOL aircraft (70+ person team, 2nd of 6 at CUASC), directing implementation of the team's first tilt-rotor VTOL system as an alternative to the 4-motor-plus-pusher layout; bench-tested and preparing for flight trials.
- Supervise 6-8 concurrent projects (tilt rotors, airdrop, gimbal, landing gear, bay doors, imaging test platform) through four formal design reviews, coordinating mechanical interfaces with the electrical and software subteams.

Structures and Payloads Mechanical Member

- Evaluated landing gear concepts on weight, compliance, and manufacturability, then designed, fabricated, and validated a carbon-fiber leaf-spring gear using SolidWorks and Ansys FEA; verified a 36% weight reduction across physical drop trials (straight, tilted, sustained load) at 1 m/s landing velocity and 15° roll.
- Characterized off-the-shelf strut suspension mechanics by executing static compression testing across incremental loads to extract effective spring stiffness, applying strain energy conservation models to predict peak dynamic deflection and boundary limits under free-fall conditions, giving a conservative upper limit safety margin.
- Reworked a DJI drone airframe to integrate custom hardware mounts and reinforce structurally unsound elements within constrained central-bay geometry, collaborating with electrical and imaging subteams to resolve interface and wiring constraints.

INTUITIVE SURGICAL **June 2026 – August 2026**

Manufacturing Process Engineering Intern

- Engineered and deployed a custom manufacturing line fixture to resolve an active bottleneck, confirmed to cut per-unit build time by 55% (40 seconds saved per unit) via line time studies.
- Executed a 132-run DoE and capability analysis in Minitab, using Cpk metrics and Pareto charts to isolate primary failure modes and reduce future testing scope by 50%.
- Designed and modeled custom machining fixtures, producing shop-ready technical drawings to support fabrication
- Managed 5 ECOs and created and managed up to 20 parts and documents across Agile PLM and Windchill, maintaining CAD version control under regulated QMS standards.

SPACE SYSTEMS DESIGN STUDIO (SSDS) **January 2026 – July 2026**

ASCENT Mechanical Subteam Member

- Precision-machined a rocket bulkhead on a manual mill and lathe, holding ± 0.005 in tolerances to ensure structural interface alignment, and fabricated custom 3D-printed mounting brackets.
- Designed a compact four-bar linkage in Fusion 360 to synchronize dual valve actuation through a 90° sweep with a single servo for liquid bi-propellant rocket, resolving engine-bay clearance constraints and preventing component interference; the linkage survived 2 hot fires and a live takeoff.

ADVANCED SPACE TECHNOLOGY RESEARCH AND ARCHITECTURES LAB **June 2025 – August 2025**

Undergraduate Researcher

- Ran 18 Density Functional Theory simulations in CP2K and 8 molecular dynamics simulations in LAMMPS, compiling binding-energy data for 16 ionic liquids to support a machine-learning model for predicting electric-propulsion fuel properties.

SKILLS

Fabrication: Manual Mill, Manual Lathe, Wet Carbon-Fiber Layups, 3D Printing, Rapid Prototyping

Software: SolidWorks, Fusion 360, Ansys Mechanical, Python, MATLAB, Java, Agile PLM, Windchill, Confluence